

Core A' — the nineteen phases, in the order spec §11 puts them

8 merged, 11 to go · 68 of 150 tasks ticked · as of 2026-09-10 · the third line of each chip is what the phase puts on fig 1d

the framework
no syn3A biology at all — D0's order, and every phase merged so far — 5b last, and it is the one that unblocks the fan-out below

✓ phase 0 · #41

Retire OpenSpec, keep its findings

— nothing in the figure —

merged 2026-09-03 · 6/6 tasks

✓ phase 1 · #42

A module contributes to a state it does not own

the mass and currency arrows inside the ODE grid

merged 2026-09-03 · 7/7 tasks

✓ phase 2 · #43

Several jump modules compose

the CME block being three coexisting boxes at all

merged 2026-09-03 · 7/7 tasks

✓ phase 3 · #44

The 1 s handshake, on a two-module toy. The kill phase

the HOOK device · catalytic and deferred-counter arrows

merged 2026-09-05 · 9/9 tasks

✓ phase 4 · #45

The 60 s rebuild

the REBUILD device · the rate-constant arrows

merged 2026-09-05 · 7/7 tasks

✓ phase 5 · #46

Growth and volume

the GROWTH device · the volume arrow

merged 2026-09-05 · 7/7 tasks

✓ phase 5b · #48

The inbound volume channel

the volume arrow's inbound half — geometry into a rate law

merged 2026-09-09 · 11/11 tasks

the fan-out — ODE block
6, 7 and 8 are mutually independent; 9 waits on 8

✓ phase 6 · #50

Central glycolysis

box: Central (10 rxns)

merged 2026-09-10 · 7/7 tasks

□ phase 7

Phosphotransferase transport and lactate export

box: Transport (6 rxns)

7/7 tasks, not merged

□ phase 8

Nucleotide recycling

box: Nucleotide (5 rxns)

8 tasks, none ticked

□ phase 10

Transcription

box: transcription (17 genes)

8 tasks, none ticked

□ phase 9

Lumped tRNA charging, in the ODE block

box: tRNA charging (the box that moved)

10 tasks, none ticked

□ phase 11

Translation

box: translation (17, plus ptsG translocation)

11 tasks, none ticked

□ phase 12

Transcript decay

box: mRNA decay (17)

8 tasks, none ticked

□ phase 13

Assemble Core A' and assert structural completeness

the whole figure asserted complete, as an object in code

8 tasks, none ticked

□ phase 14

The validation checks, in the scoping note's order

conservation checks over the figure

10 tasks, none ticked

□ phase 15

Observables and synthetic data

— nothing in the figure —

8 tasks, none ticked

□ phase 16

Recovery, coverage, and the reference

— nothing in the figure —

11 tasks, none ticked

□ phase 17

Per-module calibration

— nothing in the figure —

tasks are written once phase 16 has run

Every merged phase is framework, and that is D0 on purpose — the protocol changes first, then the kill risk on a two-module toy, before a line of syn3A biology. It is also why fig 1d looks untouched: the phases that draw boxes all lie ahead. Phases 15–17 put nothing on fig 1d at all. They are the layer the project exists for, and a state graph cannot show it.

the fan-out — CME block
10 before 11 and 12, which both read the transcripts 10 owns

assembly, validation, inference
strictly serial — each check catches what the next would mask

the inbound volume channel task 7.2 needs

charging contributes to the pools recycling owns

can proceed on a double; R1 records the cost